Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

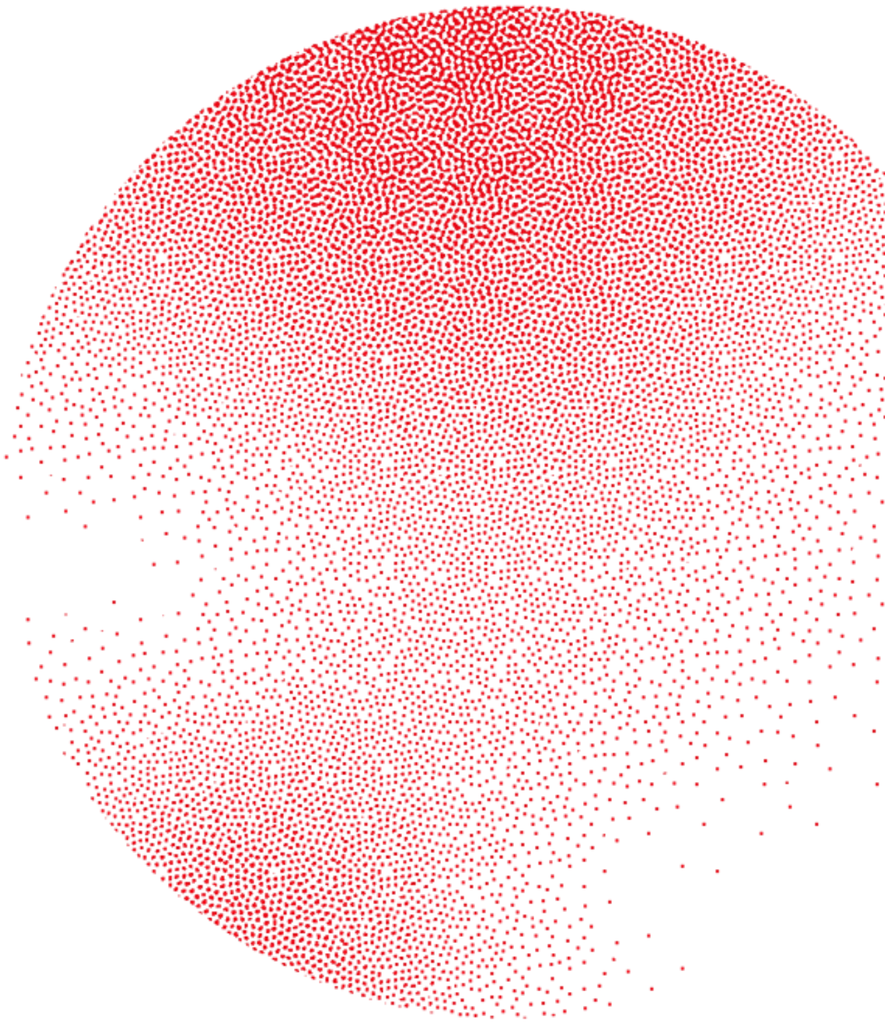
Day 2 - Session 2

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University of Fribourg
October 27th, 2026

elixir SWITZERLAND

This presentation

- Learning outcomes
- Downstream analysis
- Rarefaction curves
- Diversity
- Ordination plots
- Differential abundance
- Other data representation
- Summary



Learning outcomes

- Process Illumina paired-end 16S amplicon (eDNA) reads with the DADA2 pipeline: quality assessment, filtering, error correction, merging, chimera removal and taxonomy assignment.

- Assess the sequencing effort with rarefaction curves and estimate α-diversity (Shannon, Simpson, Chao1).

- Explore β-diversity with an NMDS ordination and barplots of the most abundant taxa.

Downstream analyses with amplicon data

From ASV/OTU tables to ecological and statistical insight

1 Exploration and QC

- Absolute abundance
- Relative abundance
- Rank abundance
- Rarefaction curves

2 Community analysis

- α-, β- and γ-diversity
- Ordination
- Multivariate analysis
- Machine learning

Complementary

3 Differential analysis

- Differentially abundant ASVs/OTUs
- LEfSe (Linear discriminant analysis Effect Size)

Various tools and platforms available

With R

main packages: phyloseq, microbiome, microbiomeSeq, DADA2, yingtools2, vegan, biomformat

Standalone

MEGAN6, micca, Mothur, QIIME2, PICRUSt2, Phinch, MetaPhlAn4

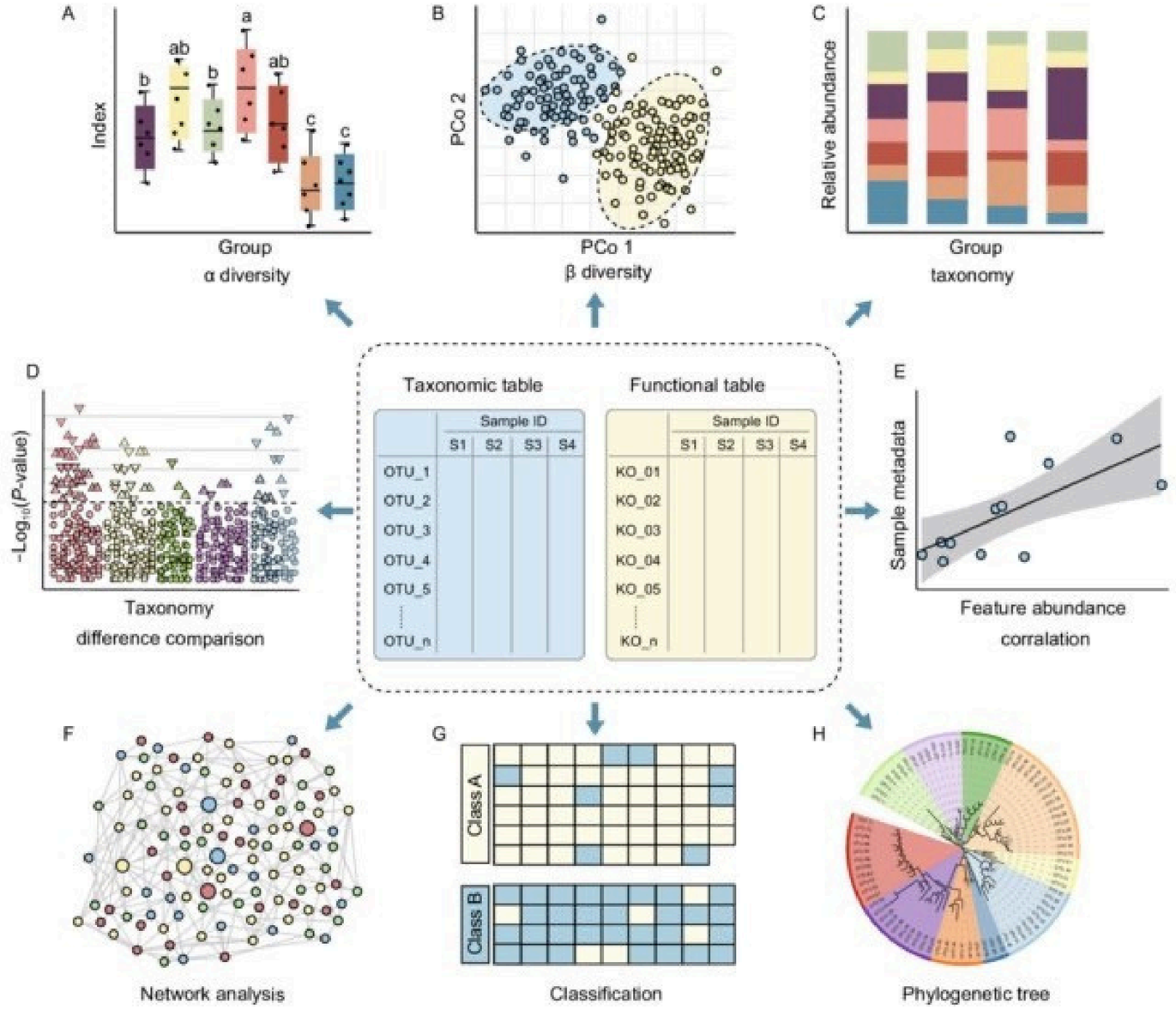
Web based

https://www.microbiomeanalyst.ca/

https://www.ebi.ac.uk/metagenomics/

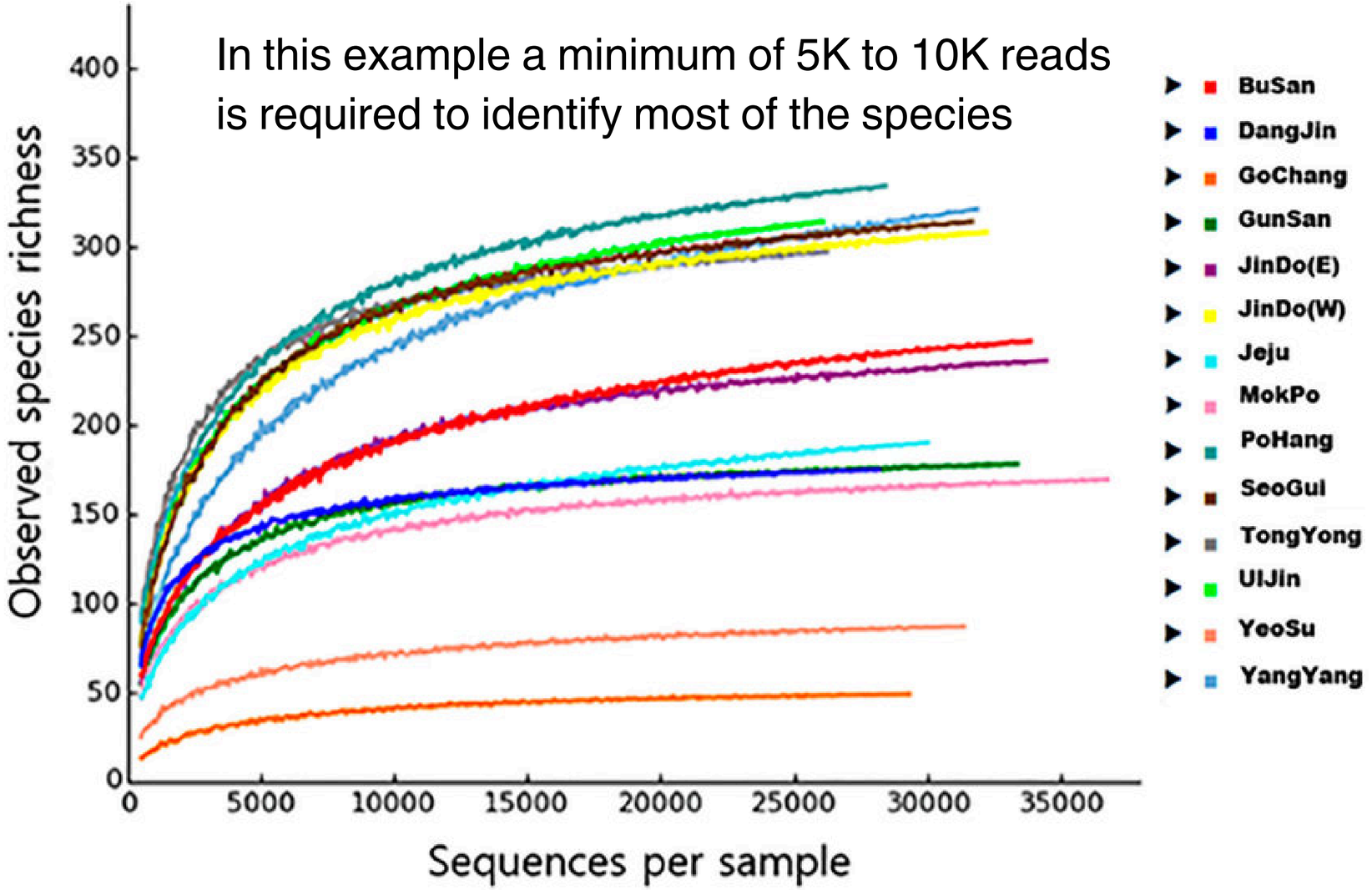
https://vamps2.mbl.edu/

Diverse representations



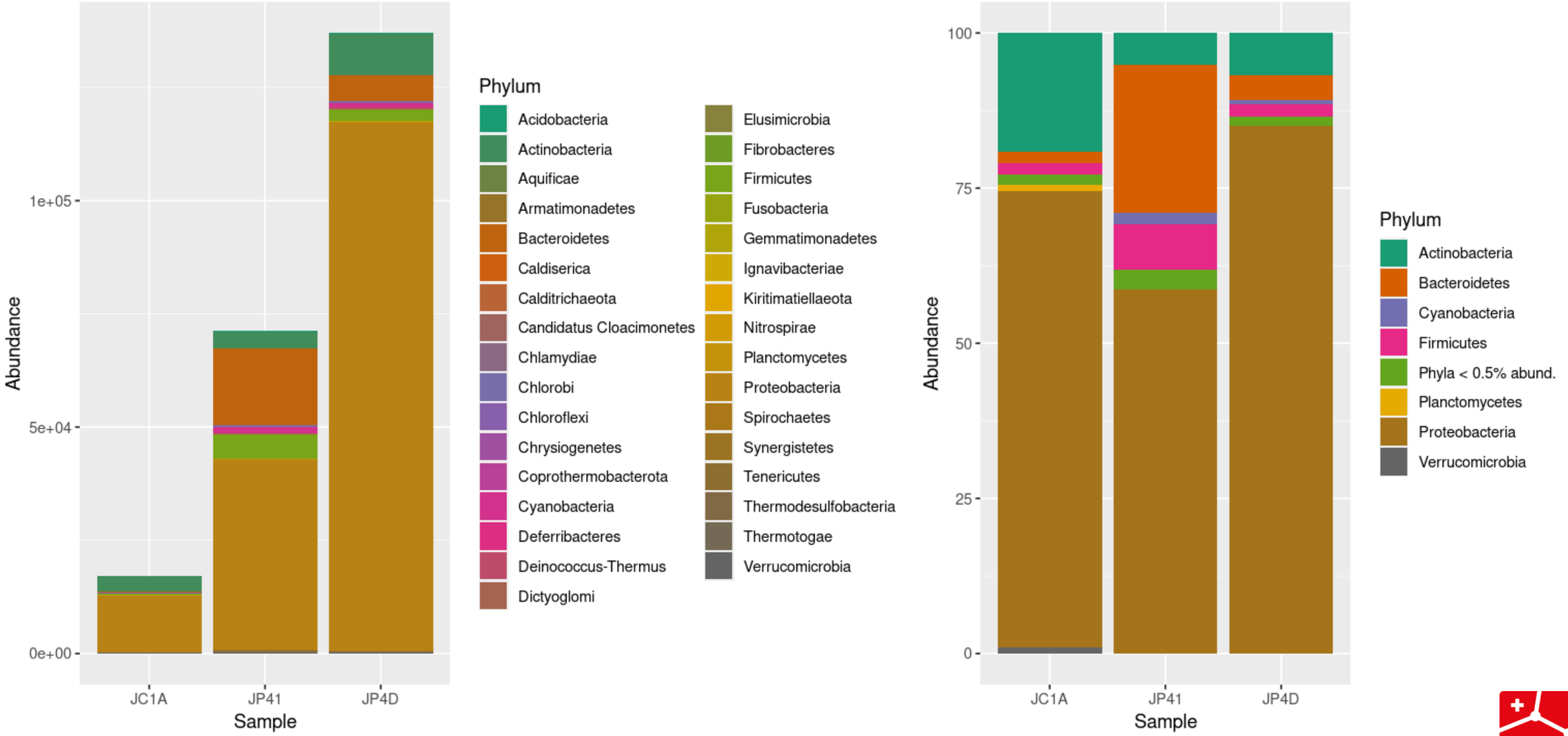
Rarefaction curves (Controversial method)

In this example a minimum of 5K to 10K reads is required to identify most of the species



Kim et al. (2016)

Absolute vs relative abundance plots (Be careful)



Diversity

Contributed diversity

α - diversity inside an area or ecosystem (species richness)

β - diversity among ecosystems

γ - diversity overall diversity of all ecosystems in a particular area

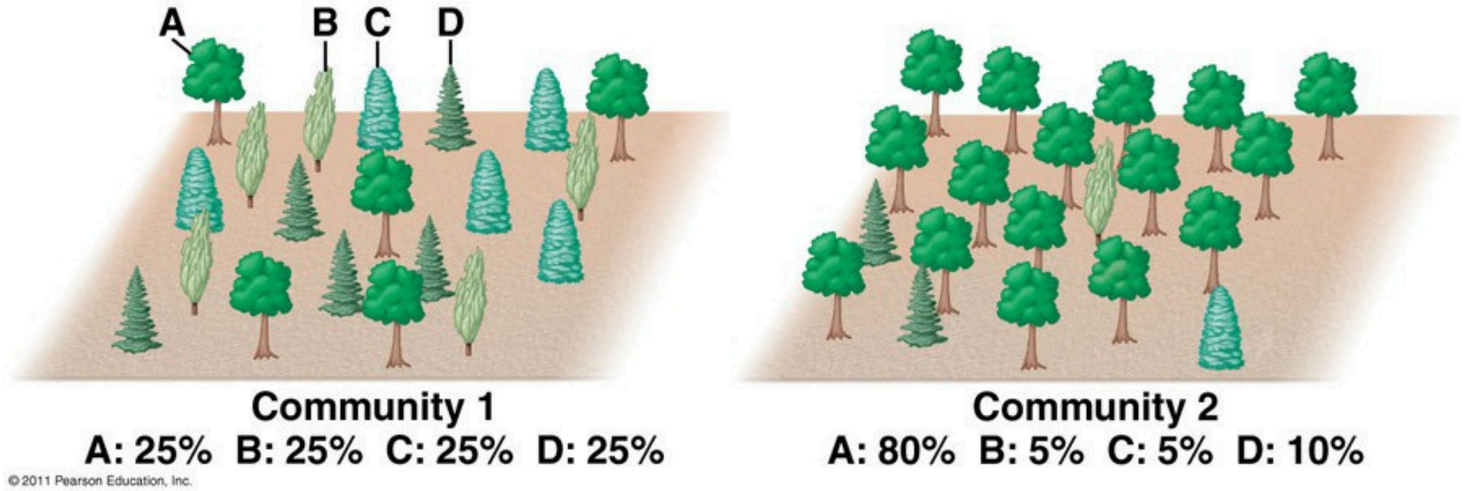
Diversity can be measured with different indexes, e.g.:

- Shannon entropy
- Inverse Simpson
- Chao1
- Count of species (but depends on sampling depth, to be checked using rarefaction curves)

Richness vs diversity

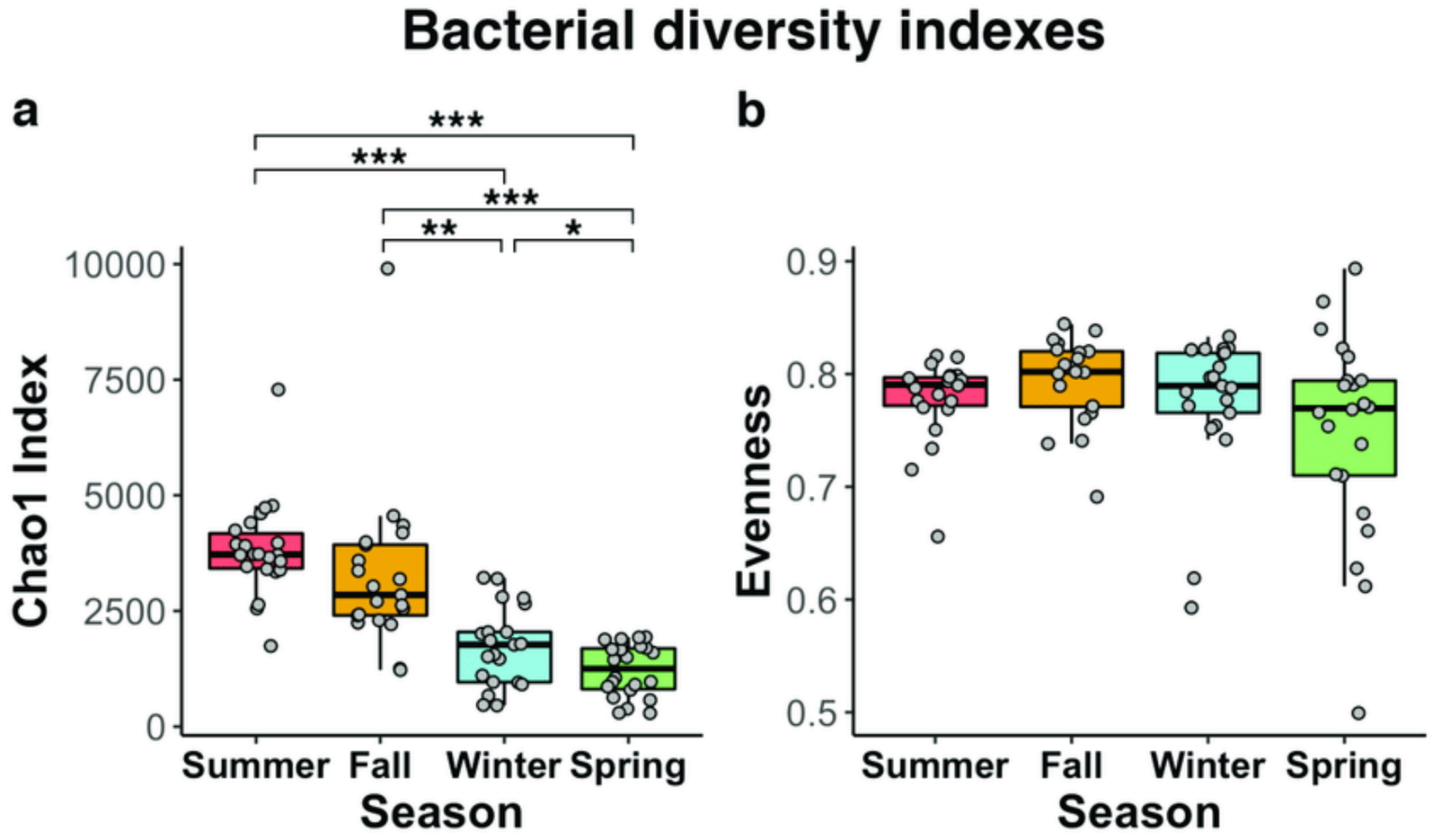
Richness: Number of observed species (OTUs/ASVs)

Diversity: Number of observed species (OTUs/ASVs) scaled to evenness



Same richness, different diversity

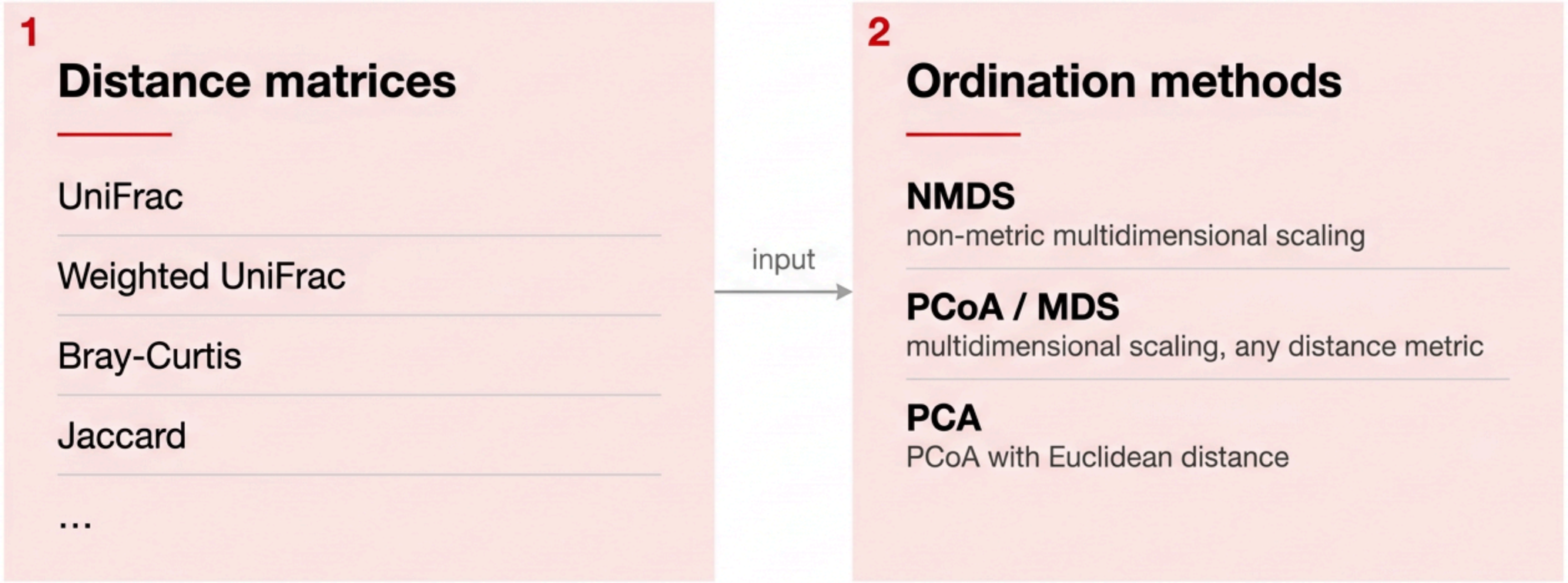
Always check with different indices



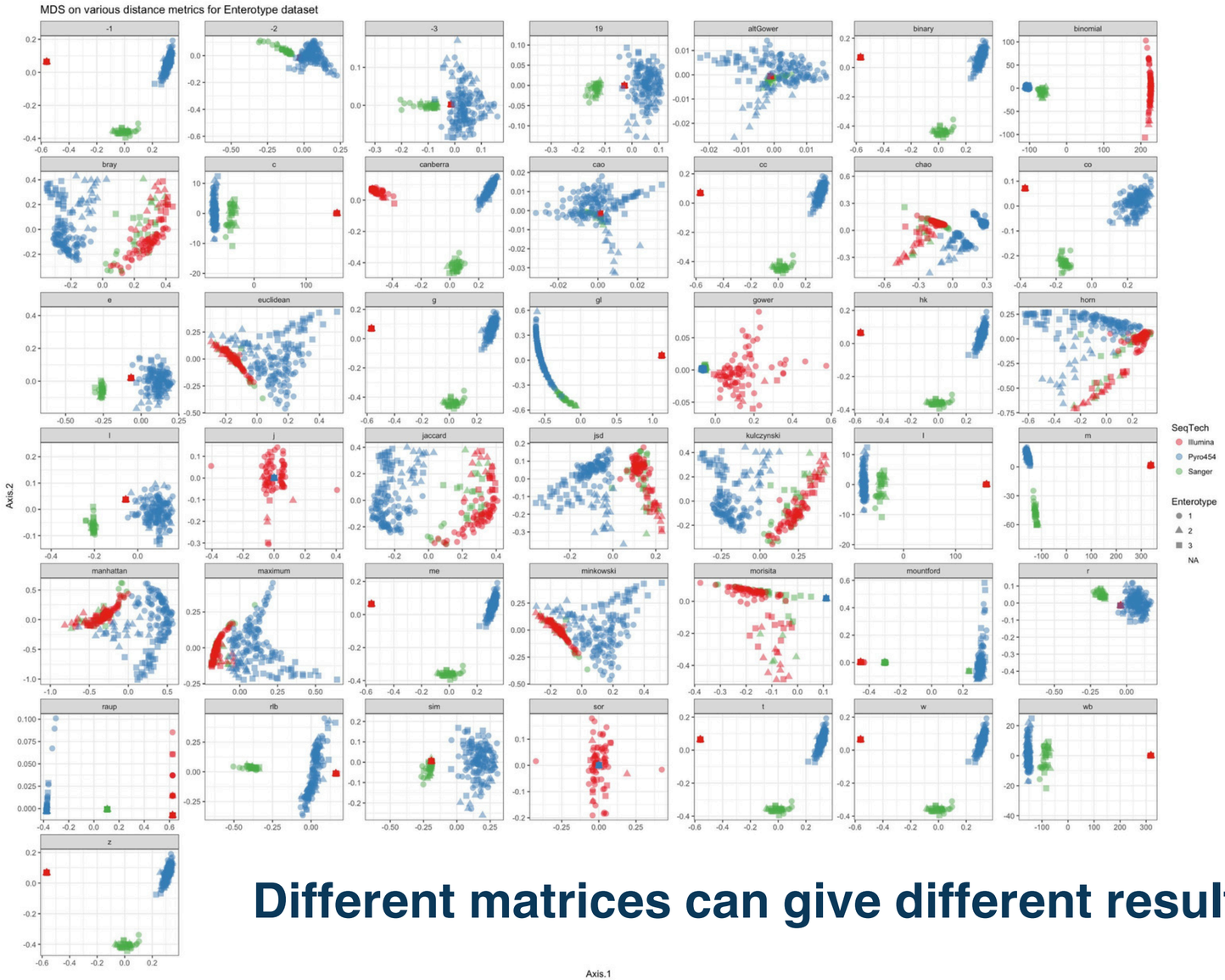
Núñez et al. (2020)

Ordination methods and distances

Ordination methods are unsupervised exploratory methods based on distance matrices and aim at dimensionality reduction.

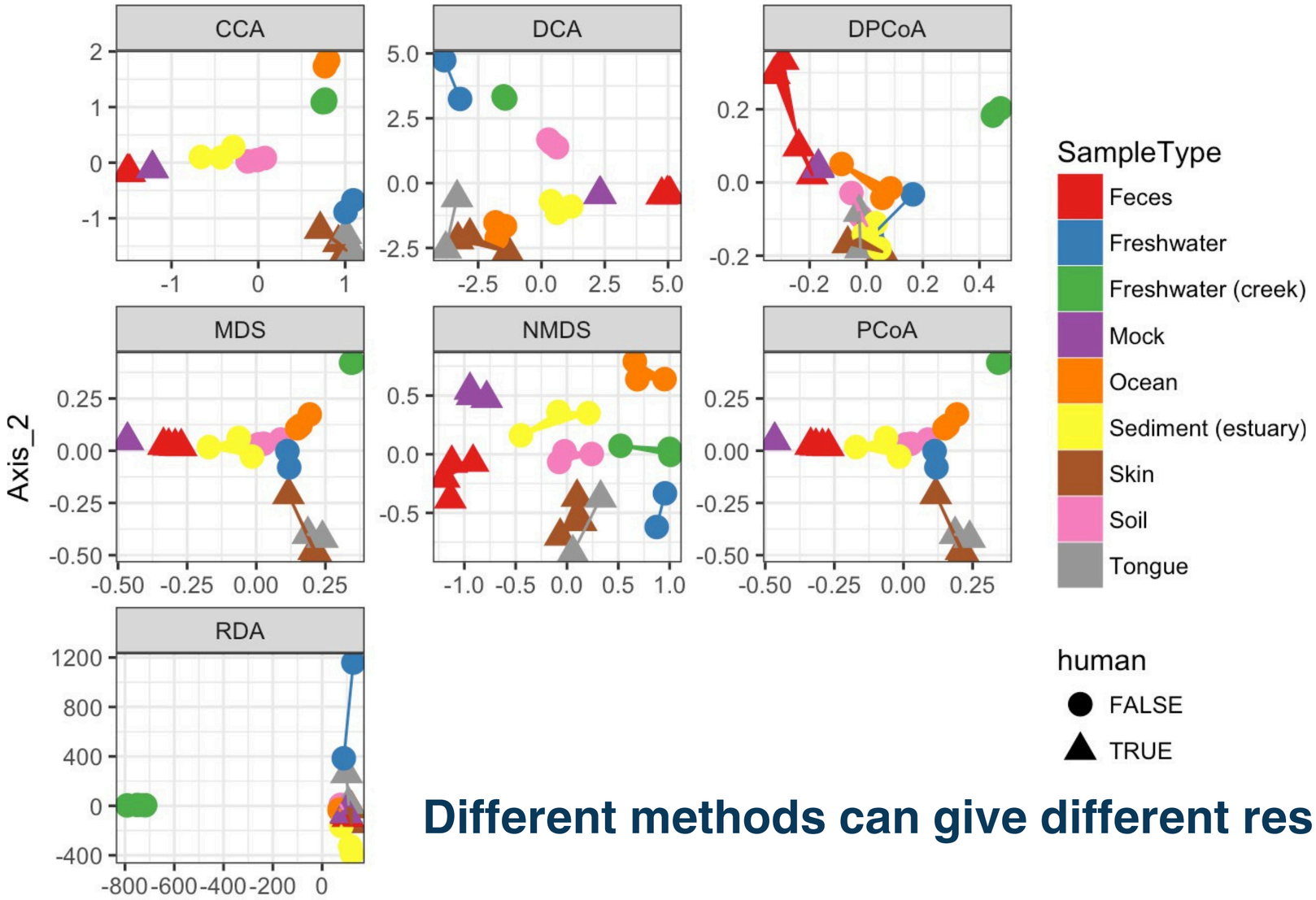


MDS on multiple distance matrices (phyloseq)



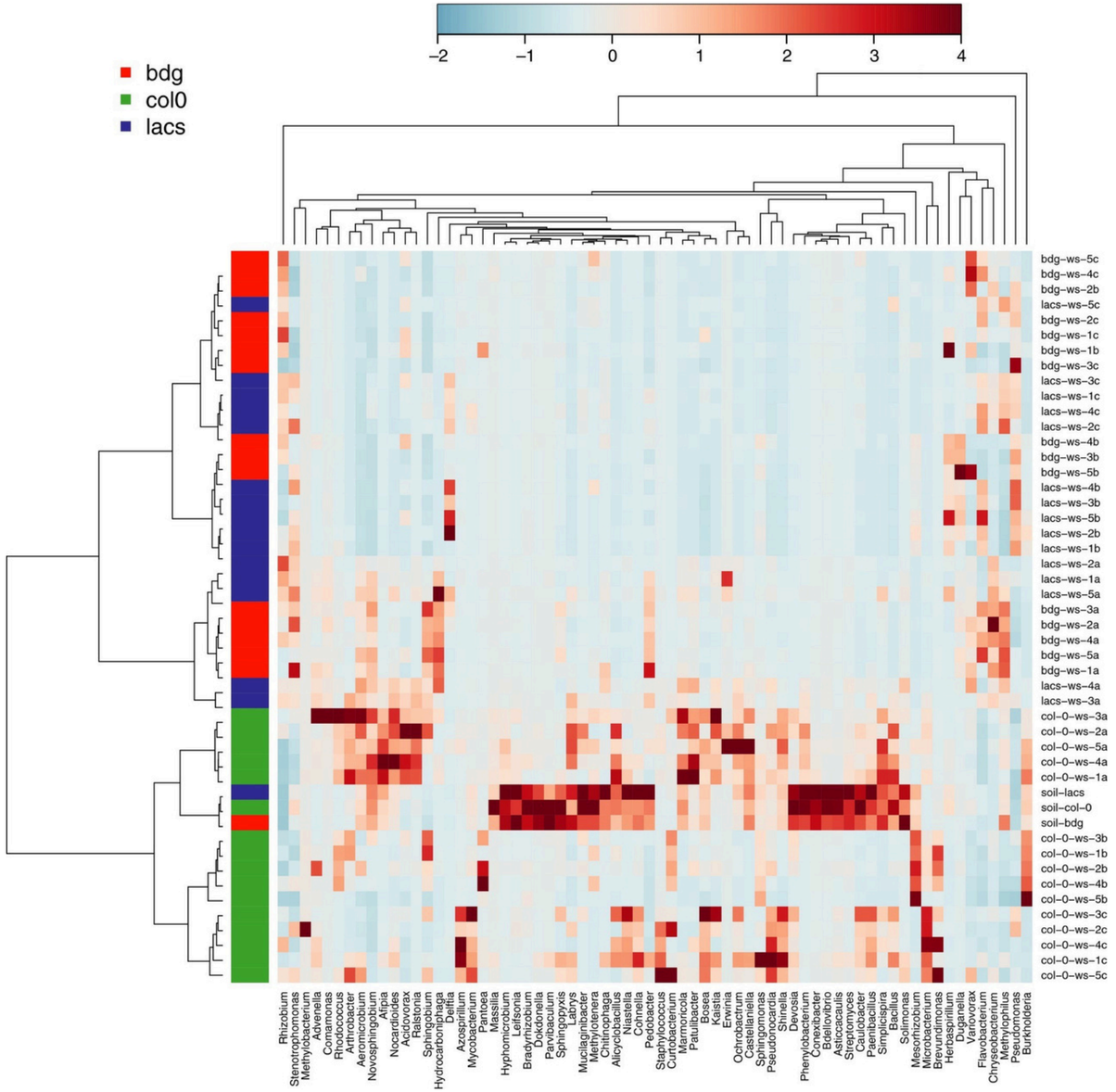
Different matrices can give different results!

Bray-Curtis distance on multiple ordination methods



Different methods can give different results!

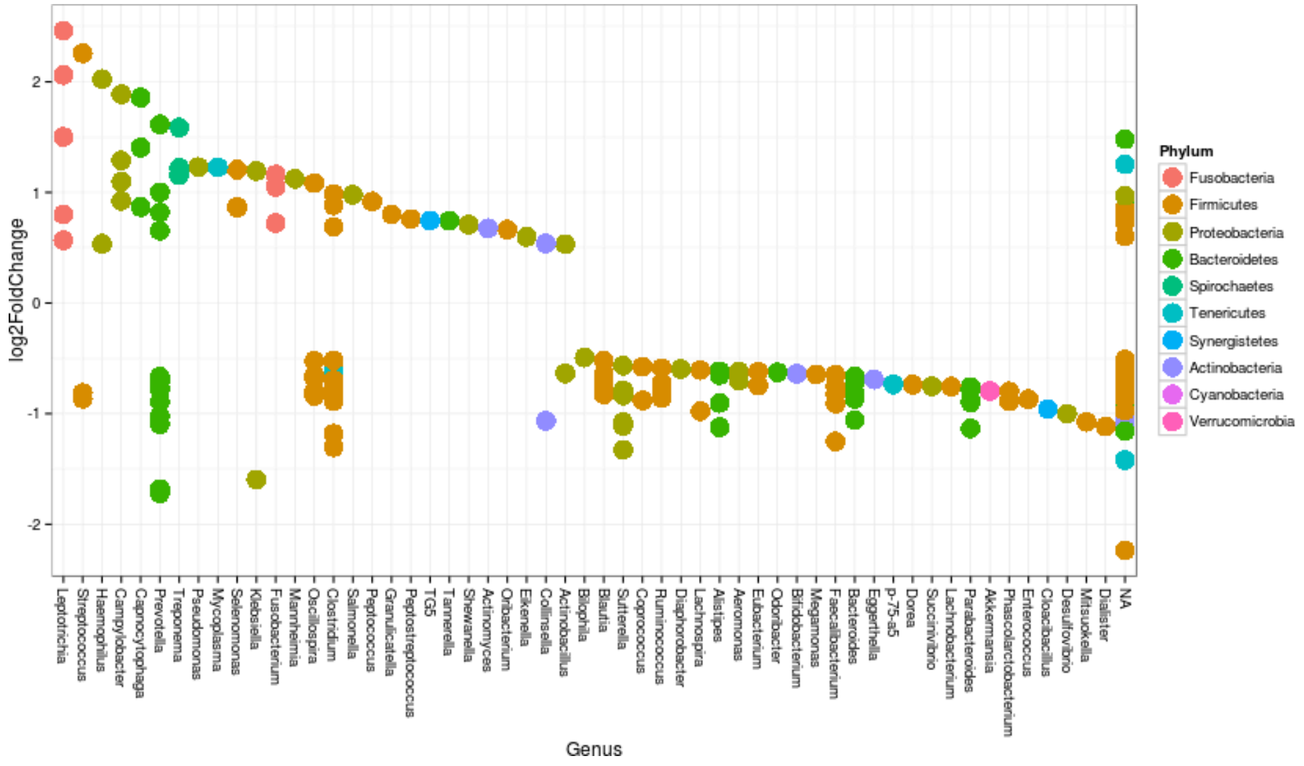
Heatmap and clustering



Differential abundance of OTUs /ASVs using phyloseq extensions

You can try to identify the differences in OTUs or ASV between pairs of samples (e.g., cancer *vs* normal)

Both DESeq2 and EdgeR are packages from the RNAseq analysis domain traditionally used with microbiome data. Not anymore!



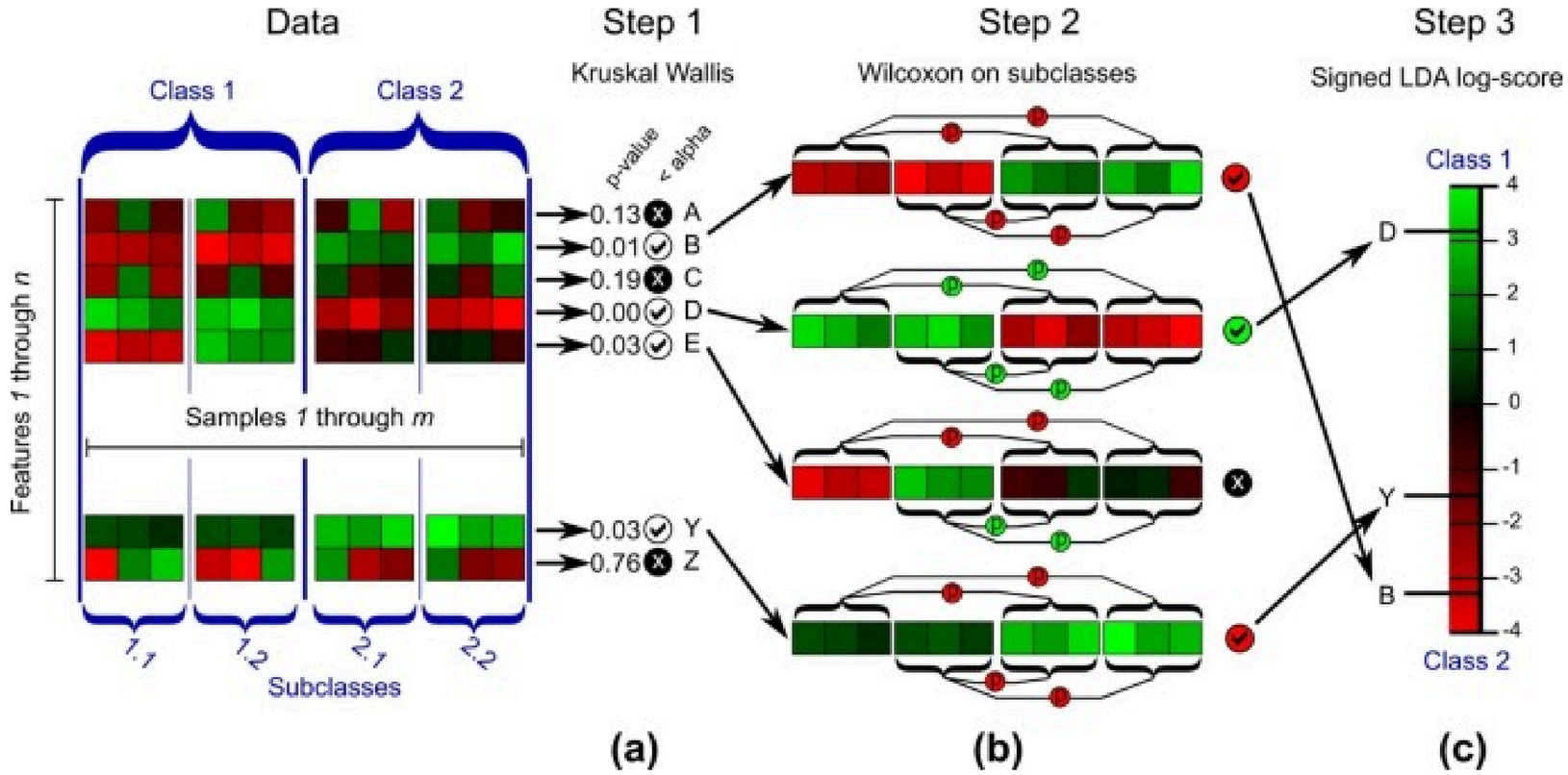
Differential abundance of OTUs /ASVs

It gives you a table similar to DEG, but for OTUs/ASVs

```
##        baseMean log2FoldChange  lfcSE   stat    pvalue      padj  Kingdom
## 304309    1.686        -0.6409 0.1823 -3.516 4.380e-04 0.0018912 Bacteria
## 16076    13.805        -0.8583 0.2094 -4.099 4.144e-05 0.0002678 Bacteria
## 561483    1.866        -0.6382 0.1809 -3.529 4.178e-04 0.0018193 Bacteria
## 177005    8.598        -0.7467 0.1905 -3.920 8.865e-05 0.0004876 Bacteria
## 469778    9.049        -0.6633 0.2209 -3.003 2.673e-03 0.0080976 Bacteria
## 308873    4.165        -0.5360 0.1794 -2.988 2.809e-03 0.0084662 Bacteria
##                Phylum                   Class             Order
## 304309      Firmicutes              Clostridia     Clostridiales
## 16076       Firmicutes              Clostridia     Clostridiales
## 561483 Actinobacteria Actinobacteria (class) Bifidobacteriales
## 177005      Firmicutes              Clostridia     Clostridiales
## 469778   Bacteroidetes             Bacteroidia     Bacteroidales
## 308873      Firmicutes              Clostridia     Clostridiales
##                   Family           Genus                   Species
## 304309     Lachnospiraceae         Blautia         Blautia producta
## 16076      Ruminococcaceae    Ruminococcus      Ruminococcus bromii
## 561483 Bifidobacteriaceae Bifidobacterium   Bifidobacterium longum
## 177005     Lachnospiraceae         Blautia                    <NA>
## 469778      Bacteroidaceae     Bacteroides  Bacteroides coprophilus
## 308873     Ruminococcaceae     Clostridium Clostridium orbiscindens
```

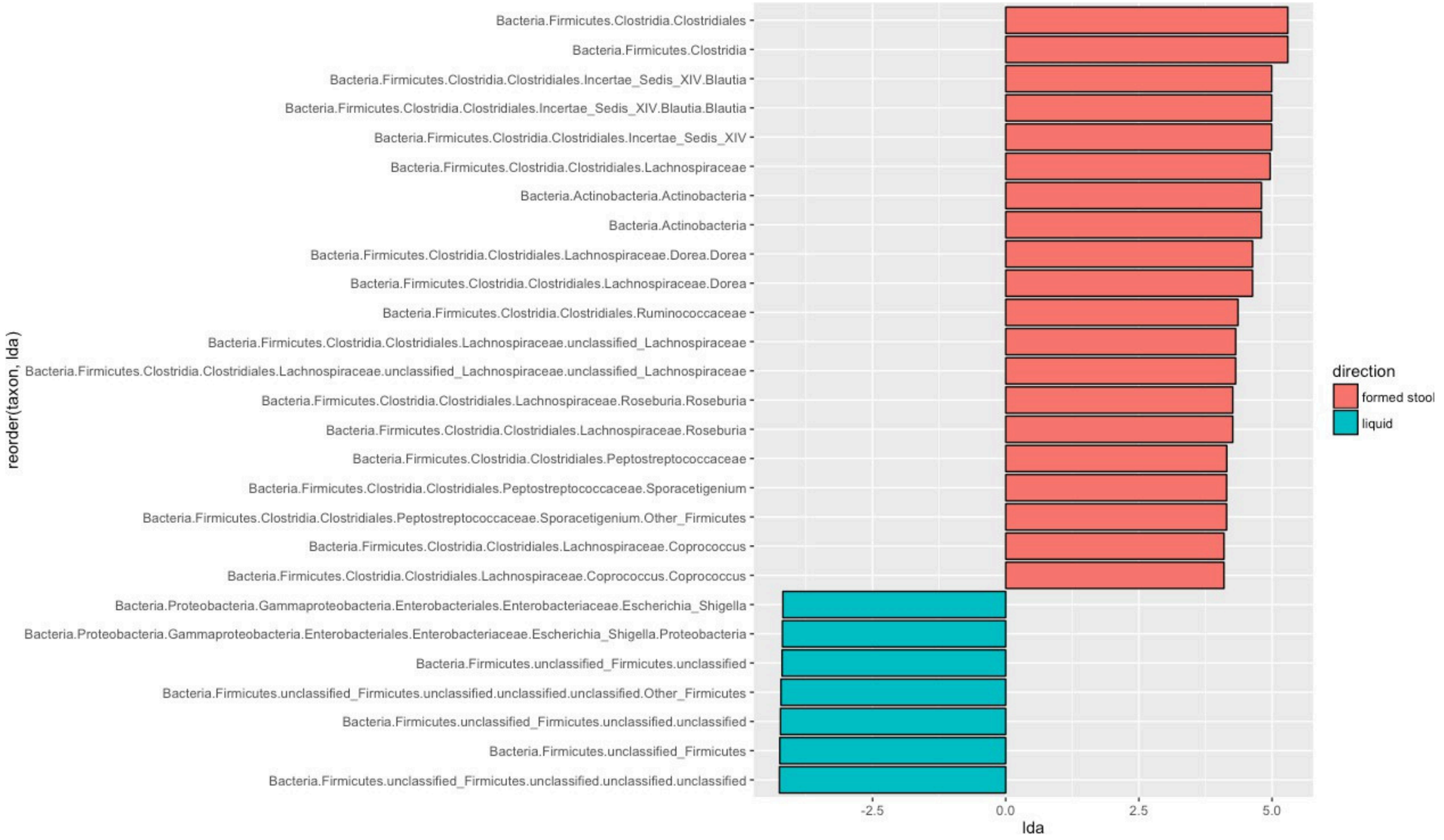
Linear discriminant Effect Size analysis (LEfSe)

Determines the features most likely to explain differences between classes by coupling standard tests for statistical significance with additional tests encoding biological consistency and effect relevance.

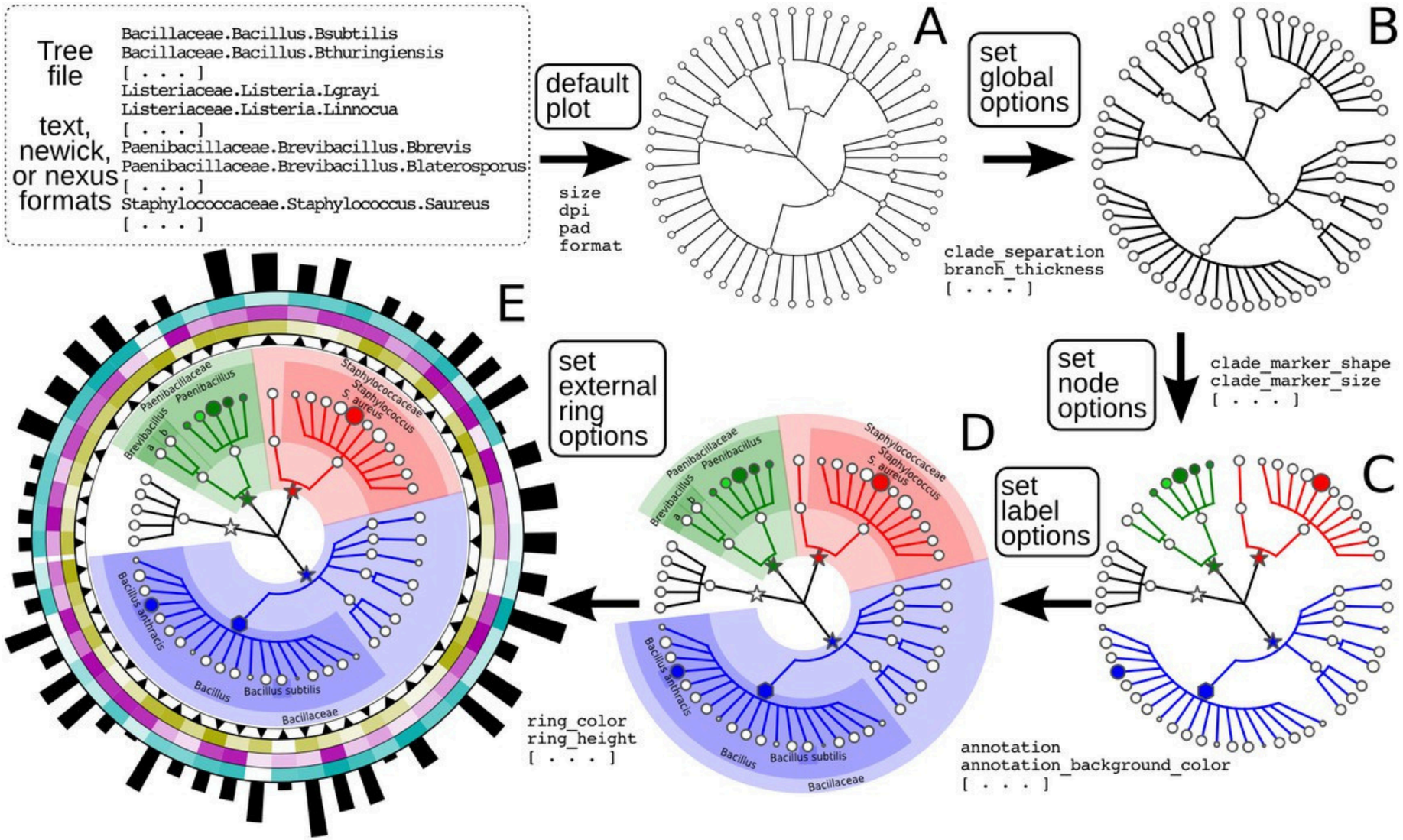


Being features (organisms, clades, operational taxonomic units, genes, or functions)

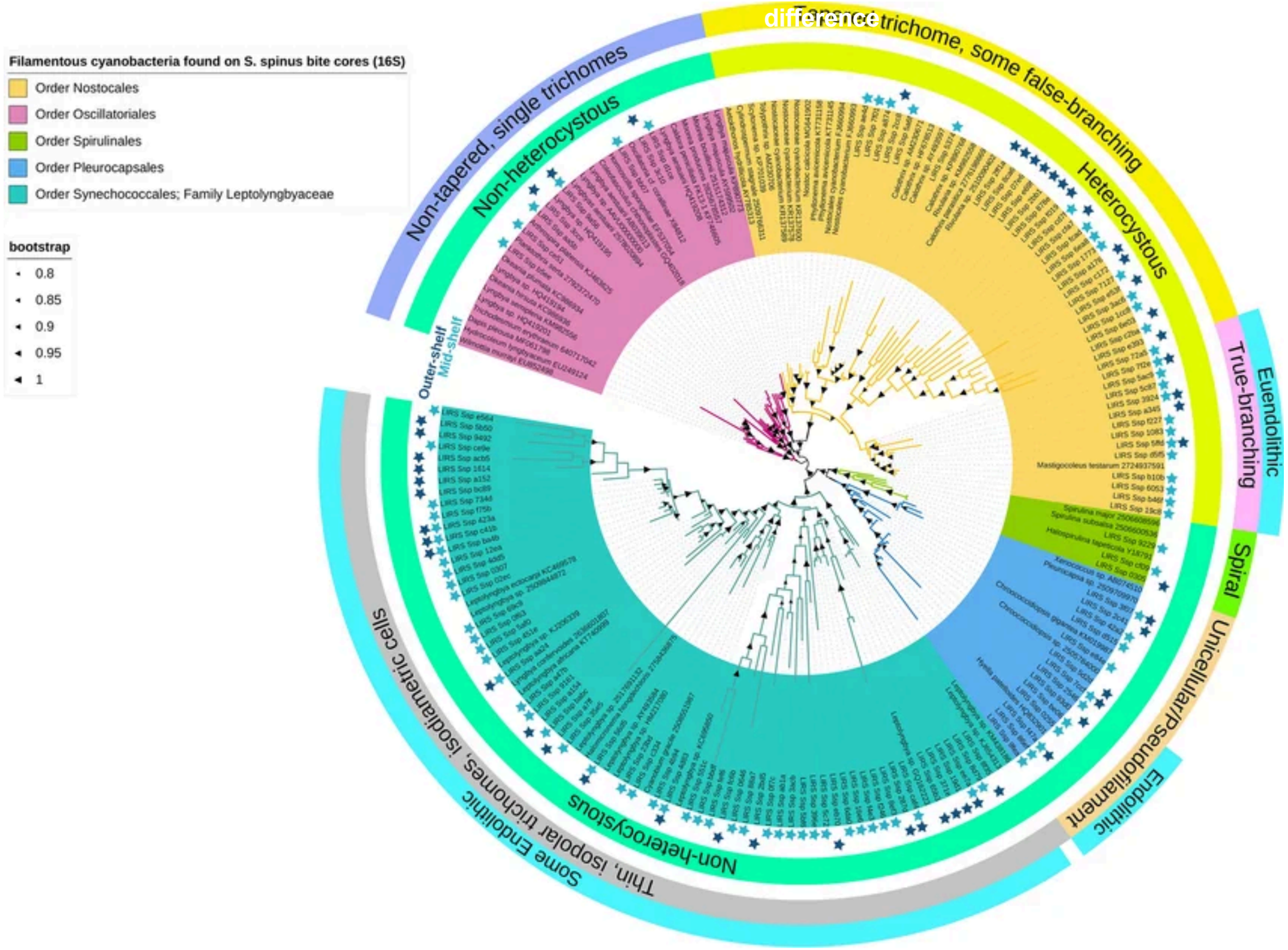
Linear discriminant Effect Size analysis (LEfSe)



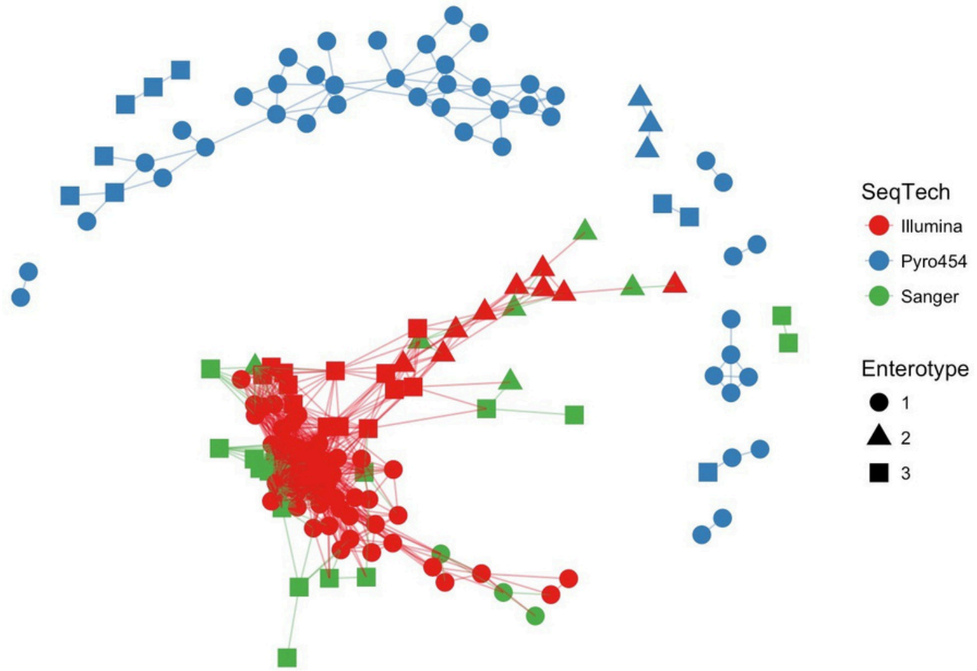
Other data representation (GraPhlAn)



Other data representation (iTOL)

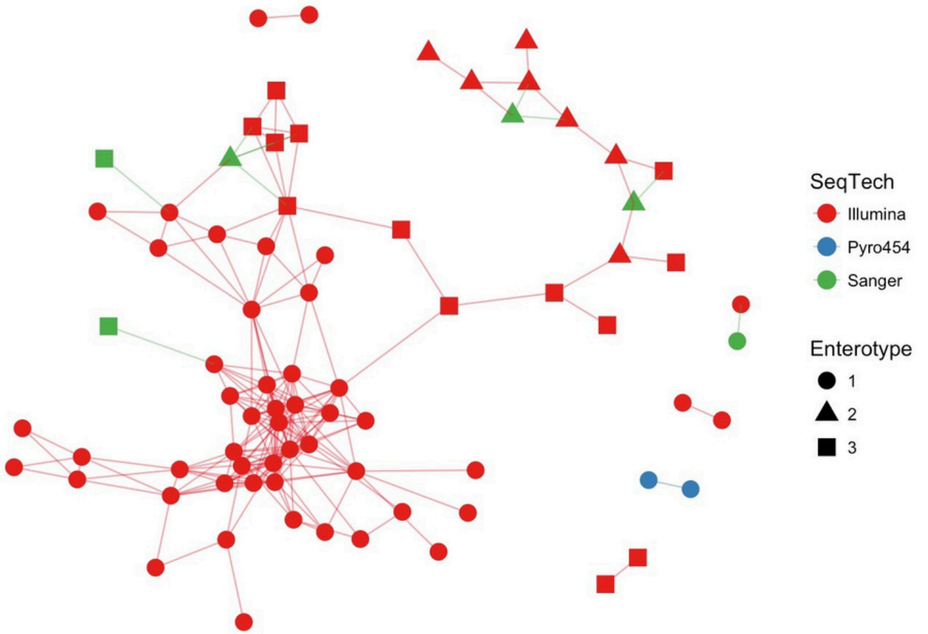


Other data representation (Network analysis)



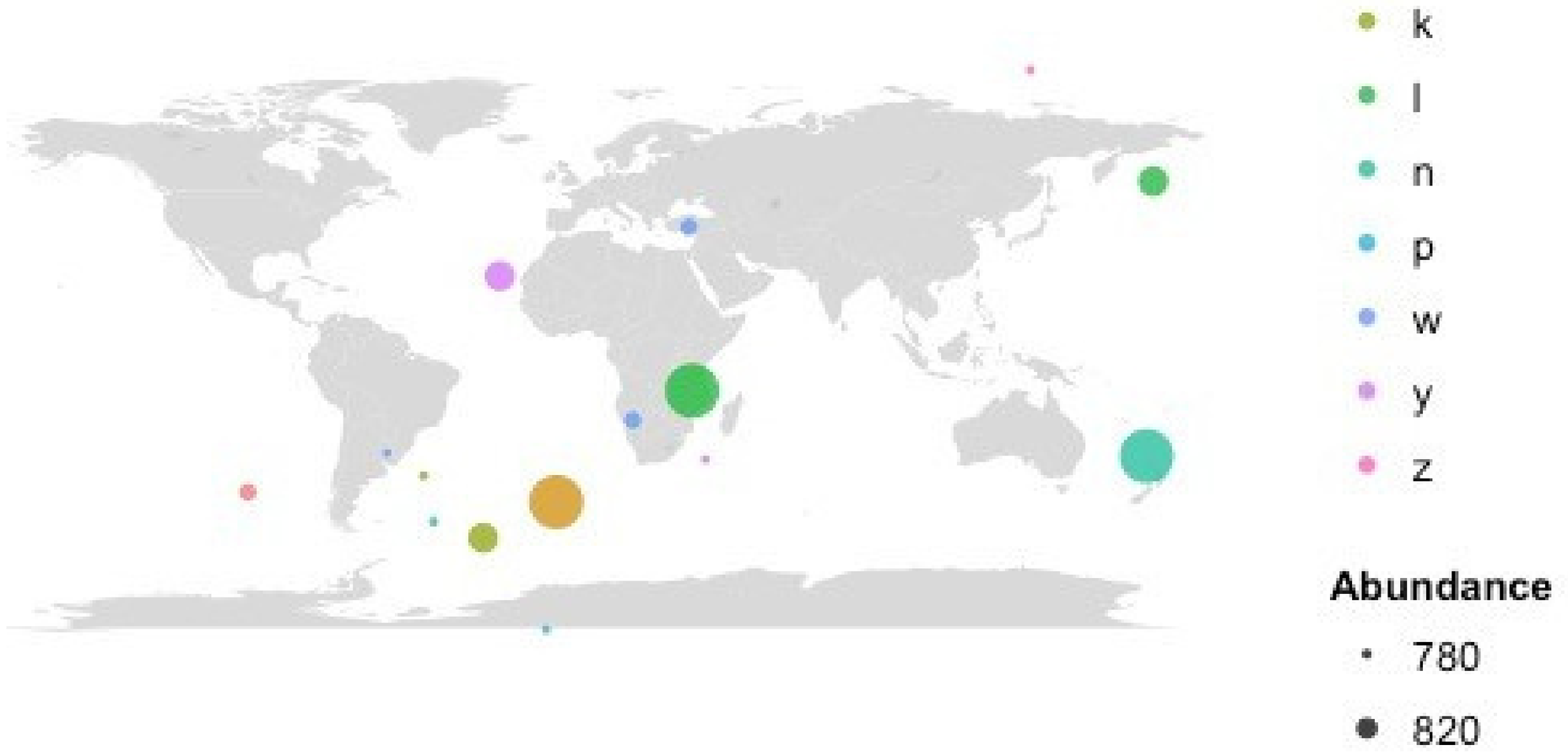
Maximum-distance
default 0.3

Same with Maximum-distance
lowered to 0.2



Other data representation (Leaflet)

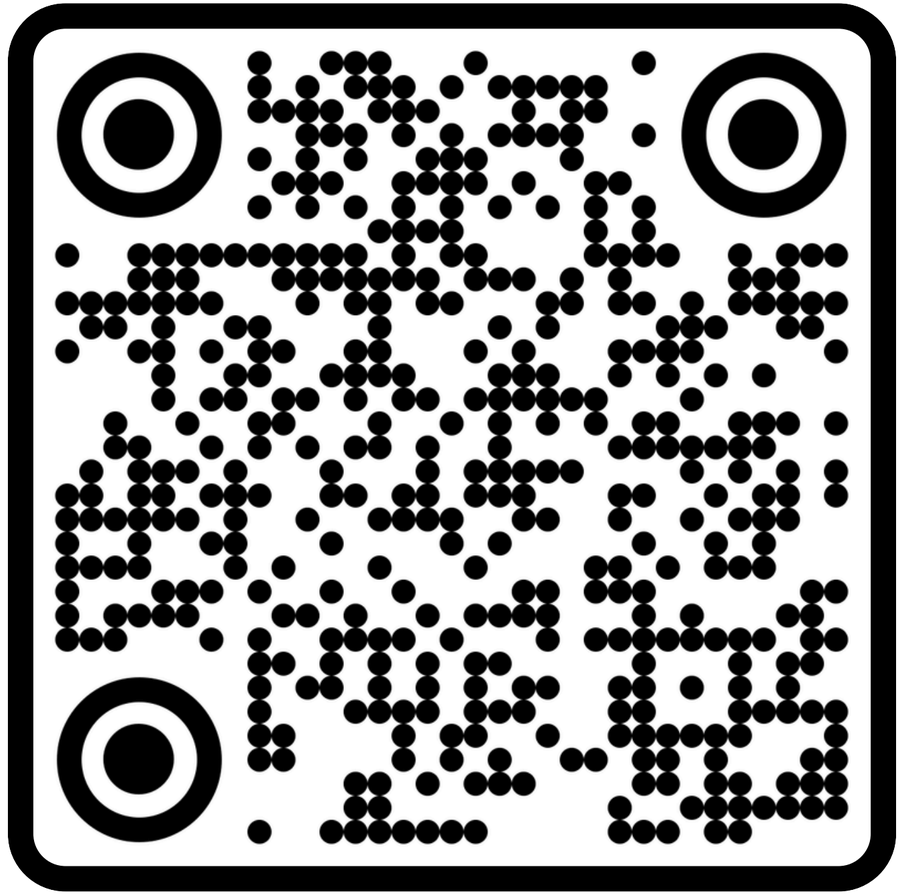
If you have metadata containing geographic coordinates, Leaflet can now be used to map a phyloseq object on a map.

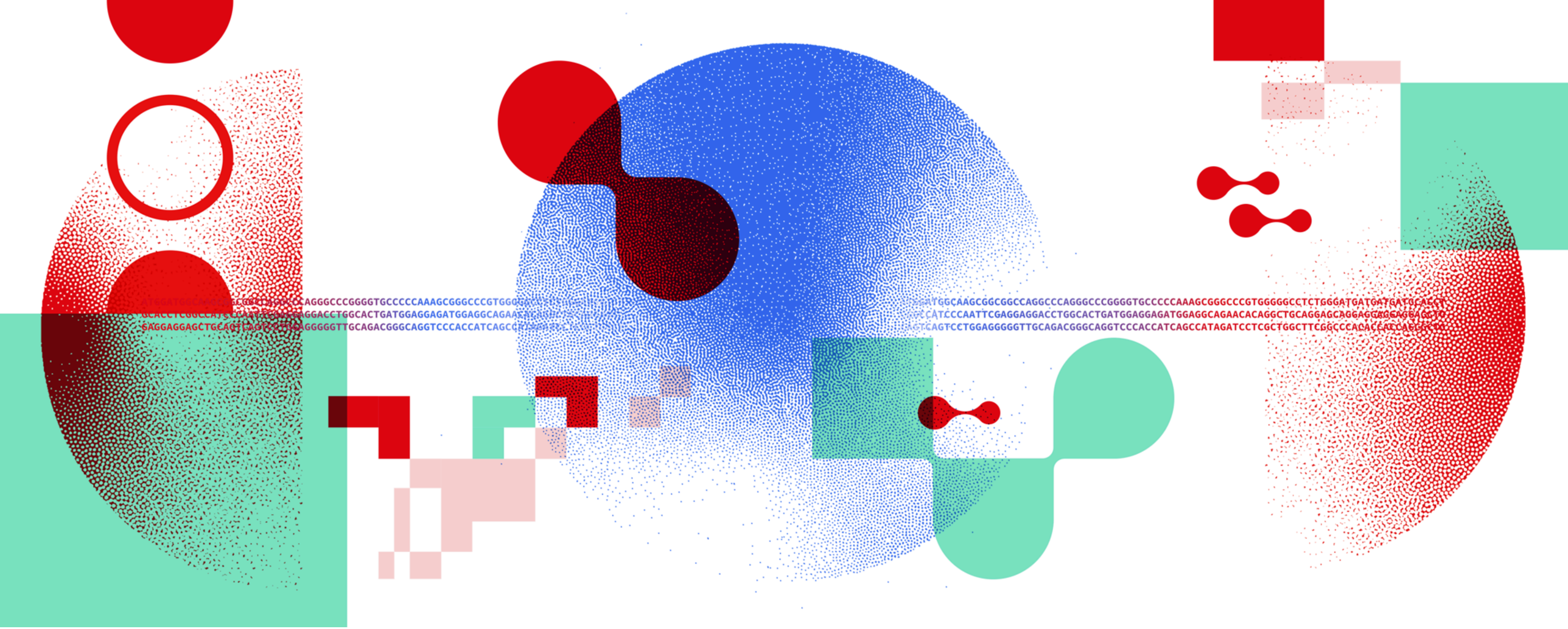


Summary

- Always start with exploratory analysis before running to evaluate significant differences.
- Use as many metrics as possible to try to cover the data structure as much as possible.
- Choose the tool that best suits the question you are trying to answer.
- Diversify the way in which you present your data.

Practicals





Thank you

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